

- 1 (a) equatorial orbit / above equator B1
 satellite moves from west to east / same direction as Earth spins B1
 period is 24 hours / same period as spinning of Earth B1 [3]
(allow 1 mark for 'appears to be stationary/overhead' if none of above marks scored)

- (b) gravitational force provides/is the centripetal force B1
 $GMm/R^2 = mR\omega^2$ or $GMm/R^2 = mv^2/R$ M1
 $\omega = 2\pi / T$ or $v = 2\pi R / T$ or clear substitution M1
 clear working to give $R^3 = (GMT^2 / 4\pi^2)$ A1 [4]

- (c) $R^3 = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (24 \times 3600)^2 / 4\pi^2$ C1
 $= 7.57 \times 10^{22}$ C1
 $R = 4.2 \times 10^7 \text{ m}$ A1 [3]
(missing out 3600 gives $1.8 \times 10^5 \text{ m}$ and scores 2/3 marks)